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Benchmarking the accuracy, robustness and reproducibility of spatial domain identification methods

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Abstract

Spatial domain identification methods are usually compared by small differences in adjusted Rand index (ARI), mostly on one tissue and one run. We benchmarked 13 published methods and three training-free baselines on 24 annotated sections of brain, breast cancer and mouse embryo from 10x Visium, STARmap, MERFISH and Stereo-seq, separating each learned embedding from its clustering step. With the same clustering for every embedding, a training-free neighbourhood-smoothing baseline had a higher mean ARI than 8 of 16 comparators and could not be separated from the other 8 (exploratory comparisons); BASS and the baseline were the most likely to rank first. Its lead came from brain tissue: on breast cancer and embryo most methods, including PCA without smoothing, were slightly ahead. On fixed embeddings, random restarts of the same Gaussian mixture changed ARI by 0.16 on average (up to 0.60), and R mclust's default fit depended on its random seed. Fit likelihood tracked ARI only on cortex. HiSTaR's published accuracy was not reproduced, and released code of one method selects its reported epoch using the labels. We recommend reporting the clustering protocol and its variance, never selecting runs with labels, and evaluating training-free baselines on several tissues.

Key words: spatial transcriptomics, spatial domains, clustering, reproducibility, benchmarking

Introduction

Spatial transcriptomics measures gene expression at thousands of locations in a tissue section while retaining their coordinates. Identifying *spatial domains*, regions with coherent expression such as cortical layers, is one of the first analyses of such data, and many methods have been proposed for it, from Bayesian and probabilistic models [1; 2], spatially aware dimension reduction [3] and neighbourhood-augmented features [4; 5; 6] to graph neural networks [7; 8; 9] and hierarchical graph variational autoencoders such as HiSTaR [10].

Most of these methods are evaluated on the twelve manually annotated human dorsolateral prefrontal cortex (DLPFC) sections [11] and report median adjusted Rand index (ARI) [12] improvements of a few hundredths over previous methods [8; 9; 10]. Several (for example GraphST, STAGATE and SEDR) share the same last step: the learned embedding is clustered once with the EEE Gaussian mixture of the R package mclust [13]. Benchmarks have found that rankings depend strongly on the dataset [14; 15], that deep methods vary with the random seed [16], that replacing each method's clustering algorithm changes results [15], and that preprocessing and clustering choices often matter more than architectural novelty [17]. Simple training-free pipelines can also be competitive [18; 19]. What these studies do not measure directly is how much of the variation comes from the clustering step itself when the embedding is held fixed.

We benchmark 13 published methods and three training-free baselines along three axes. *Accuracy*: how do the methods compare across four technologies and three tissue types when their embeddings are clustered in the same way, and how does this change with each method's own clustering? *Robustness*: on a *fixed* embedding, how much does ARI change with the random initialisation of the mixture model and with the random seed of R mclust, and does selecting fits by likelihood help? *Reproducibility*: do published results reproduce with the released code (here HiSTaR's DLPFC results), and does released code select models using the labels? The study design is summarised in Figure 1. We find that a standard neighbourhood-smoothing baseline is among the most accurate methods on brain tissue but not on the other tissues, that the clustering step varies more than most differences between methods, and that released code can use the labels for model selection.

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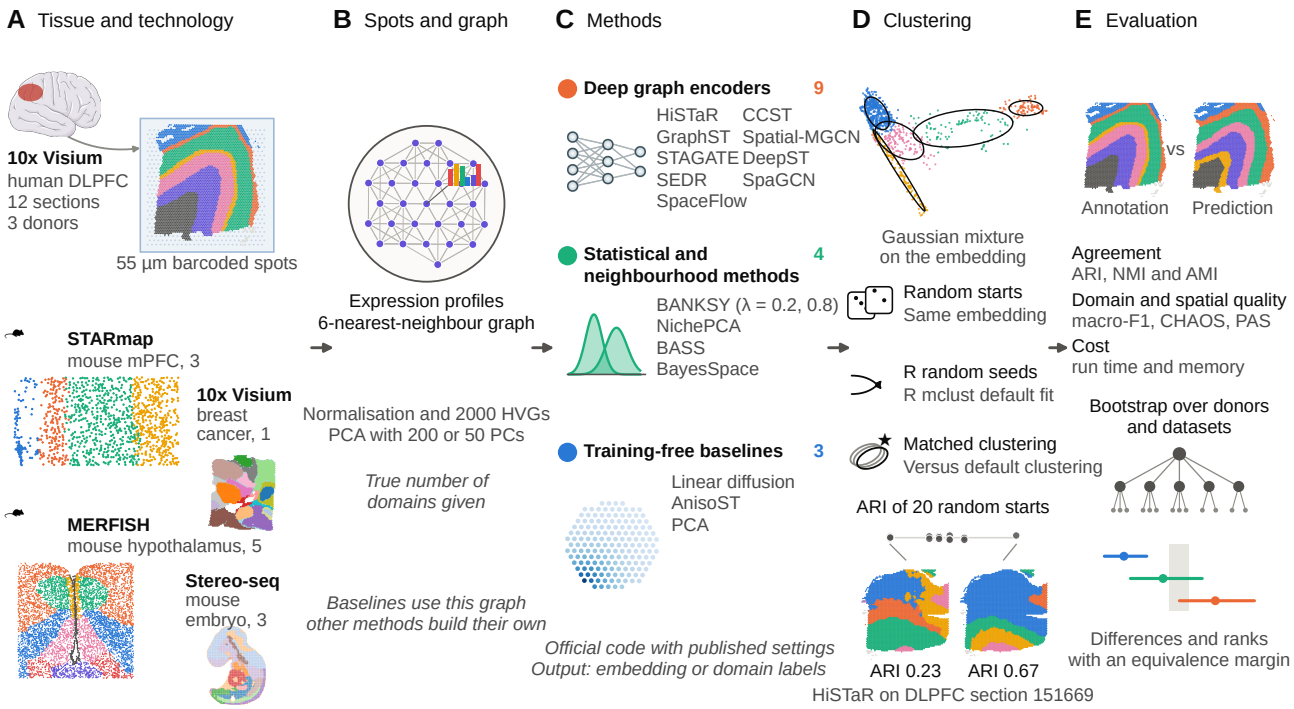


Fig. 1. Study design. (**A**) Examples of the data: human DLPFC on a 10x Visium capture area (section 151673) and mouse sections profiled by STARmap (mPFC, section BZ5), MERFISH (hypothalamus, Bregma −0.14), human breast cancer on 10x Visium and a mouse embryo profiled by Stereo-seq (E9.5); numbers give the sections per dataset; colours are the annotation, used only for scoring. (**B**) Each spot or cell carries an expression profile; the training-free baselines use the six-nearest-neighbour graph, and other methods build their own graph as documented. The true number of domains is given to every method. (**C**) 13 published methods and three training-free baselines in three families (family colours are used in Figures 1, 2 and 5). (**D**) Each embedding is clustered with the method's default clustering and with a matched clustering (the same 20-start Gaussian mixture for every method); single-start refits and R mclust with different random seeds measure the variance of the clustering step. Below: one fixed HiSTaR embedding of DLPFC section 151669 refitted from 20 random starts (ARI of each start) and the domains of the worst and best start. (**E**) Predicted domains are compared with the annotation using agreement, per-domain, spatial and cost metrics, summarised with a bootstrap over donors and datasets. Icons: Bioicons (CC0) and Servier Medical Art (CC BY 3.0).

Materials and Methods

Data

DLPFC (10x Visium). Twelve sections from three donors [11]; seven annotated layers (five for donor 2), `layer_guess_reordered`. **STARmap mouse medial prefrontal cortex.** Three sections (about 1050 cells, 166 genes) with four annotated layers (L1, L2/3, L5, L6) [20]. **MERFISH mouse hypothalamus.** Animal 1, five sections from Bregma −0.04 to −0.24 (about 5500 cells, 155 genes) with eight annotated regions [21]. The STARmap and MERFISH annotations are those distributed with the BASS analyses [22], one of the benchmarked methods (see Limitations). **Human breast cancer (10x Visium).** Block A Section 1 of the 10x Genomics demonstration data [23] (3798 spots), with the 20 regions (ductal and lobular carcinoma *in situ*, invasive ductal carcinoma, tumour edge and healthy tissue) of the annotation distributed with the SEDR analyses [24]. **Mouse embryo (Stereo-seq).** Three sections of the MOSTA atlas [25] at bin50 resolution (about 25 µm): E9.5 E1S1 (5913 bins, 12 annotated tissues), E10.5 E2S1 (8494 bins, 18) and E10.5 E1S1 (18,408 bins, 13), with the published tissue annotation. These two datasets were added after the analysis of the first three was complete, without changing any setting. In total, 24 sections from four technologies (Supplementary Table S1; Supplementary Note 1).

Preprocessing

DLPFC, breast cancer and Stereo-seq (whole transcriptome; Stereo-seq bins without counts removed): genes in fewer than 50 spots or with fewer than 10 counts were removed, 2000 highly variable genes were selected (Seurat v3 flavour), counts were library-size normalised, log-transformed and scaled, and reduced to 200 principal components (PCs). STARmap and MERFISH (targeted panels): all genes, same normalisation, 50 PCs. All methods received the same input where they accept one; BANKSY used the log-normalised genes, and SpaGCN, GraphST and the other published methods used their own documented preprocessing of the same raw counts. The true number of domains was given to every method, and unannotated spots were excluded from evaluation.

Methods

Methods were chosen to cover the main families of spatial domain identification with maintained public code that runs on CPU: deep graph encoders, statistical and neighbourhood methods, and Bayesian models. Every method was run with its official code and default or tutorial settings on one CPU thread; none was tuned per dataset. Versions, settings and deviations from the tutorials (for example the neighbourhood radius, expressed in spot spacings so that it transfers across technologies) are listed in Supplementary Table S2.

- **Deep graph encoders.** HiSTaR [10] (PyTorch port of the official code with the published defaults: cross-level weight 0.3, 200+200 epochs), GraphST [9] (official package and tutorial: 600 epochs, 20-PC projection, R mclust; with and without its label refinement; on Stereo-seq its documented setting for Stereo-seq data, `datatype='Stereo'`), SpaGCN [7] (tutorial without histology, Louvain resolution search to the true number of domains), STAGATE [8], SEDR [24], SpaceFlow [26], CCST [27], Spatial-MGCN [28] and DeepST [29]. Spatial-MGCN's released test script reports the epoch with the highest ARI against the annotation; in the benchmark we report its final epoch (label-free) and give the label-selected value separately.
- **Statistical and neighbourhood methods.** BANKSY [4] (official `pybanksy`, 18 neighbours, $m_{\max} = 1$, 20 PCs; $\lambda = 0.8$, recommended for domain segmentation, and $\lambda = 0.2$, recommended for cell typing, included for reference), NichePCA [18] (a published training-free method: neighbour aggregation and PCA), and the Bayesian models BASS [22] and BayesSpace [1], which cluster internally. BayesSpace requires a spot lattice and was run on the Visium sections only.
- **Training-free baselines.** *Linear diffusion* is standard neighbourhood smoothing (graph Laplacian smoothing): the top 30 PCs are smoothed on the spatial six-nearest-neighbour graph, $H^{(t+1)} = (1-\alpha)H^{(t)} + \alpha D^{-1}AH^{(t)}$, $\alpha = 0.5$, $T = 10$. *AnisoST* uses the same scheme with edge weights recomputed at every step, $w_{ij}^{(t)} = \exp(-\|h_i^{(t)} - h_j^{(t)}\|^2/\kappa^{(t)})$ (Perona–Malik-type edge-preserving diffusion [30]; $\kappa^{(t)}$ the median squared edge difference); a similar idea has been used inside a deep model [31]. *PCA* without smoothing is the non-spatial reference. All baseline settings were fixed on DLPFC donor 1 before any other data were analysed (sensitivity: Supplementary Figure S2).

Seeds: three per section for most methods; one for STAGATE, CCST, Spatial-MGCN and BASS in the published pipelines and for GraphST on DLPFC, and one run per section for BayesSpace (Supplementary Table S23). The breast cancer and Stereo-seq sections were run only in the standardised runner described below, with three seeds per method (two for CCST on the largest Stereo-seq section, where one run takes more than 5 h; one run for BayesSpace). On these two datasets jobs ran one at a time with a memory cap of 12 GB on a 15 GB machine; the earlier runs on the first three datasets had no cap (the largest, BayesSpace on DLPFC, used 12.5 GB). BASS, GraphST, NichePCA, Spatial-MGCN and DeepST exceeded the cap on the largest Stereo-seq section (18,408 bins), and DeepST seeds 1 and 2 on E10.5 E2S1; these 17 jobs were rerun with the same code and settings on a second server with 125 GB RAM (CPU only, a different processor). Their accuracy and peak memory are reported with the rest; their run times are not compared with the others. For GraphST, the call to R mclust was wrapped for the newer software stack of that server; it gave identical labels on test embeddings (Supplementary Note 6).

Neutrality.

The authors developed AnisoST, and linear diffusion and PCA are its ablations; the authors did not develop any of the published methods. To limit bias, all baseline settings were frozen on the four sections of DLPFC donor 1, every published method was run with its authors' defaults, and all comparisons are repeated without the six development sections (those four and the first STARmap and MERFISH sections, used to test the pipelines).

Clustering protocols

Default clustering: the clustering step of each method's official pipeline. This is R mclust (EEE) for GraphST, STAGATE and SEDR; a 3-start scikit-learn Gaussian mixture with tied covariance [32] for HiSTaR, BANKSY and the three baselines; Leiden or k-means for SpaceFlow, NichePCA, CCST, Spatial-MGCN and DeepST; and the internal model for SpaGCN, BASS and BayesSpace. *Matched clustering* (primary analysis): every embedding clustered with the same tied-covariance Gaussian mixture (the mclust EEE model) from 20 random starts, keeping the fit with the highest log-likelihood (label-free). SpaGCN, BASS and BayesSpace output no embedding and keep their own clustering under both protocols. *Initialisation experiment*: one seed-0 embedding per section and method, refitted 20 times with a single random start each (14 methods and baselines that output an embedding). *R mclust experiment*: R mclust 6.0.1 (`Mclust(Z, G = k, modelNames = "EEE")`) on the same embeddings, (i) with its default initialisation under ten R random seeds and fixed row order, (ii) initialised on all observations (`initialization = list(subset = 1:n)`), and (iii) the same after adding Gaussian noise with 1% of each feature's standard deviation, five times.

Evaluation and statistics

ARI against the manual annotation is the primary metric; NMI gave a nearly identical ranking of methods (Spearman $\rho = 0.97$ between method means; Supplementary Table S4), and further agreement, spatial and label-free metrics are in Supplementary Table S14 (definitions in Supplementary Note 3). For every method and section we average over seeds. Sections are not independent (the 12 DLPFC sections come from three donors; the STARmap, MERFISH and Stereo-seq sections come from one study each), so the unit of analysis is the group: 7 groups (three DLPFC donors, STARmap, MERFISH, breast cancer and Stereo-seq). Five of the seven groups are brain tissue, so the overall comparison weights brain data most; we therefore also report each dataset separately. For each method we estimate the difference in mean ARI from linear diffusion, the reference, which was chosen after the means were seen, so these comparisons are exploratory. Intervals come from a hierarchical bootstrap (5000 resamples of groups, then of sections within a group; the estimate is the equal-weight mean of group means) with 95% percentile intervals (Supplementary Note 4). A method is called equivalent to the reference if the 90% interval lies within ± 0.03 ARI, about the seed-to-seed standard deviation. Ranks and the probability of ranking first come from the same bootstrap. The analysis is repeated without the development sections (DLPFC donor 1 and the first section of STARmap and MERFISH). With 7 groups the intervals are wide; we report uncorrected 95% intervals and, as a check, Bonferroni-corrected intervals over the 16 comparisons, and interpret only consistent patterns. Run time and peak memory were measured in a standardised runner that gives every method the same input (process-tree sampling, one CPU thread per job; Supplementary Note 6); its accuracy agrees with the published pipelines to within the differences listed in Supplementary Table S16. For the first three datasets the accuracy analyses use the published pipelines; for breast cancer and Stereo-seq they use the runner, which records both the method's own clustering and matched clustering.

Results

Linear diffusion is among the most accurate methods on brain tissue

With matched clustering, linear diffusion had a mean ARI of 0.558, 0.705 and 0.497 on DLPFC, STARmap and MERFISH, 0.545 on breast cancer and 0.306 on Stereo-seq (Figure 2A; per-dataset values for both protocols in Supplementary Table S3). Across all methods and both protocols, the highest dataset means were reached by linear diffusion on DLPFC (0.558), BANKSY $\lambda = 0.8$ on STARmap (0.740), BASS on MERFISH (0.582), BASS on breast cancer (0.624) and SpaGCN on Stereo-seq (0.388). Linear diffusion had a higher mean ARI than 8 of 16 comparators (95% interval above zero; Figure 2B; Supplementary Table S6), and this held for 7 of them without the development sections. The baseline was ahead of BayesSpace, SEDR, DeepST, SpaceFlow, SpaGCN, PCA, CCST and Spatial-MGCN. The other 8 comparisons were unresolved (BASS, BANKSY $\lambda = 0.8$, AnisoST, NichePCA, STAGATE, HiSTaR, GraphST and BANKSY $\lambda = 0.2$; for example BASS Δ -0.002, 95% interval -0.059 to $+0.063$, and BANKSY $\lambda = 0.8$ $+0.024$, -0.020 to $+0.069$), and no comparator was equivalent to the baseline within ± 0.03 ARI. With a Bonferroni correction only 3 comparisons (SpaceFlow, CCST and Spatial-MGCN) remained above zero. Ranked on all 24 sections, BASS was first in 0.50 of the bootstrap resamples, linear diffusion in 0.32 and BANKSY λ=0.8 in 0.04; 6 methods had rank 1 within their 95% interval (Figure 2C; Supplementary Table S9). Taking all datasets from the standardised runner instead of the published pipelines gave a similar picture, with a different verdict only for HiSTaR (9 comparisons favouring the baseline, 7 unresolved; first rank BASS 0.40, linear diffusion 0.49; Supplementary Table S7). With each method's default clustering 11 of 16 comparisons favoured the baseline and 5 (BASS, BANKSY $\lambda = 0.8$, AnisoST, BANKSY $\lambda = 0.2$ and BayesSpace) were unresolved (Supplementary Table S5).

The lead of the baseline depends on the tissue

With matched clustering, no comparator had a higher mean ARI than linear diffusion on DLPFC, 1 on STARmap and 1 on MERFISH (BASS in both cases; BASS was also ahead in an equal-weight average of the three brain datasets, 0.594 against 0.587), but 13 of 16 comparators were ahead on breast cancer and 11 of 15 on Stereo-seq. Averaged over comparators, linear diffusion exceeded the comparators by 0.172 ARI on the three brain datasets and trailed them by 0.023 on the other two. On these two datasets the methods were close to each other (Figure 2A), and PCA without any smoothing was also ahead of linear diffusion (0.562 against 0.545 on breast cancer, 0.349 against 0.306 on Stereo-seq): smoothing with the settings fixed on DLPFC did not help on these tissues, and most methods lay within a narrow range. The breast cancer annotation has 20 numbered regions of four classes; scored against the four classes with $K = 4$, every method with an embedding reached a low ARI (0.04–0.31) and the order changed (NichePCA and SpaceFlow highest; linear diffusion 0.19; Supplementary Table S18). On the largest Stereo-seq section (18,408 bins) BASS, GraphST, NichePCA, Spatial-MGCN and DeepST needed 12–28 GB of RAM (BASS 22, NichePCA 20, GraphST 12, Spatial-MGCN 27, DeepST 28 GB), more than the 12 GB cap of the benchmark machine. Restricting the comparison to the datasets with an expert annotation (dropping STARmap, whose annotation provenance is unknown, and Stereo-seq, whose annotation was derived partly by clustering) gave the same picture: 7 comparisons favoured the baseline, 9 were unresolved, and BASS and linear diffusion ranked first in 0.36 and 0.37 of the resamples (Supplementary Table S8). Example partitions are shown in Figure 3 and, for breast cancer and Stereo-seq, in Supplementary Figure S3.

Number of domains, seed variability and cost

Every method was given the annotated number of domains K. Changing K by one changed ARI by 0.06 on average (the larger of the two directions 0.09, up to 0.49; Supplementary Table S17). BIC did not provide a usable label-free choice: within $K \pm 2$ it selected the largest value in 329 of 336 embeddings. At that choice the ranking of methods was largely preserved (Spearman $\rho = 0.91$ between method means), but BANKSY $\lambda = 0.8$ and NichePCA moved ahead of linear diffusion (0.462, rank 3), so the lead of the baseline depends on giving every method the true K.

Seed-to-seed variability differed between methods: HiSTaR's standard deviation of ARI was 0.047 on DLPFC and 0.180 on STARmap, against 0.012–0.054 for the training-free baselines (Supplementary Table S10). The median time per job on one CPU thread ranged from under 30 s for the baselines and BANKSY to about 46 min for CCST, and CCST took more than 5 h on the largest Stereo-seq section (Supplementary Table S15). The edge-preserving variant AnisoST did not improve on linear diffusion overall (MERFISH: 0.363 vs 0.497), although on DLPFC it was more accurate on spots at layer boundaries (by 0.025, 9 of 12 sections, Wilcoxon signed-rank $p = 0.034$, uncorrected; Supplementary Figure S4); at single-cell resolution neighbouring cells often differ by cell type, so an edge-preserving rule blocks the smoothing needed to reveal regions.

Random initialisation of the clustering step changes ARI on a fixed embedding

Figure 4A shows the ARI of 20 single-start fits of the same mixture model on each fixed embedding. Across the 336 embeddings of 14 methods and baselines, the ARI range within an embedding was 0.16 on average and reached 0.60 (SEDR, DLPFC section 151672). The training-free baselines were not immune: the mean range was 0.18 for linear diffusion, against 0.09 for SpaceFlow and 0.22 for BANKSY $\lambda = 0.8$ (Supplementary Table S11). The best of the 20 fits by ARI exceeded the average single fit by 0.07: choosing a clustering run by its agreement with the labels, explicitly or implicitly by trying several seeds, would inflate a result by more than the few hundredths typically reported between successive methods. Every fit for three representative methods is shown in Supplementary Figure S1.

The default fit of R mclust depends on its random seed

For data with more than 2000 observations, R mclust initialises the EM algorithm by hierarchical agglomeration on a random subset of 2000 observations drawn with R's random number generator, so its default fit depends on the R seed. The published pipelines fix this seed (2020 in STAGATE and GraphST, 2023 in SEDR), which makes their result reproducible but tied to an arbitrary choice. Across the 290 embeddings with more than 2000 spots or cells, changing only the R seed changed ARI by 0.11 on average (up to 0.46;

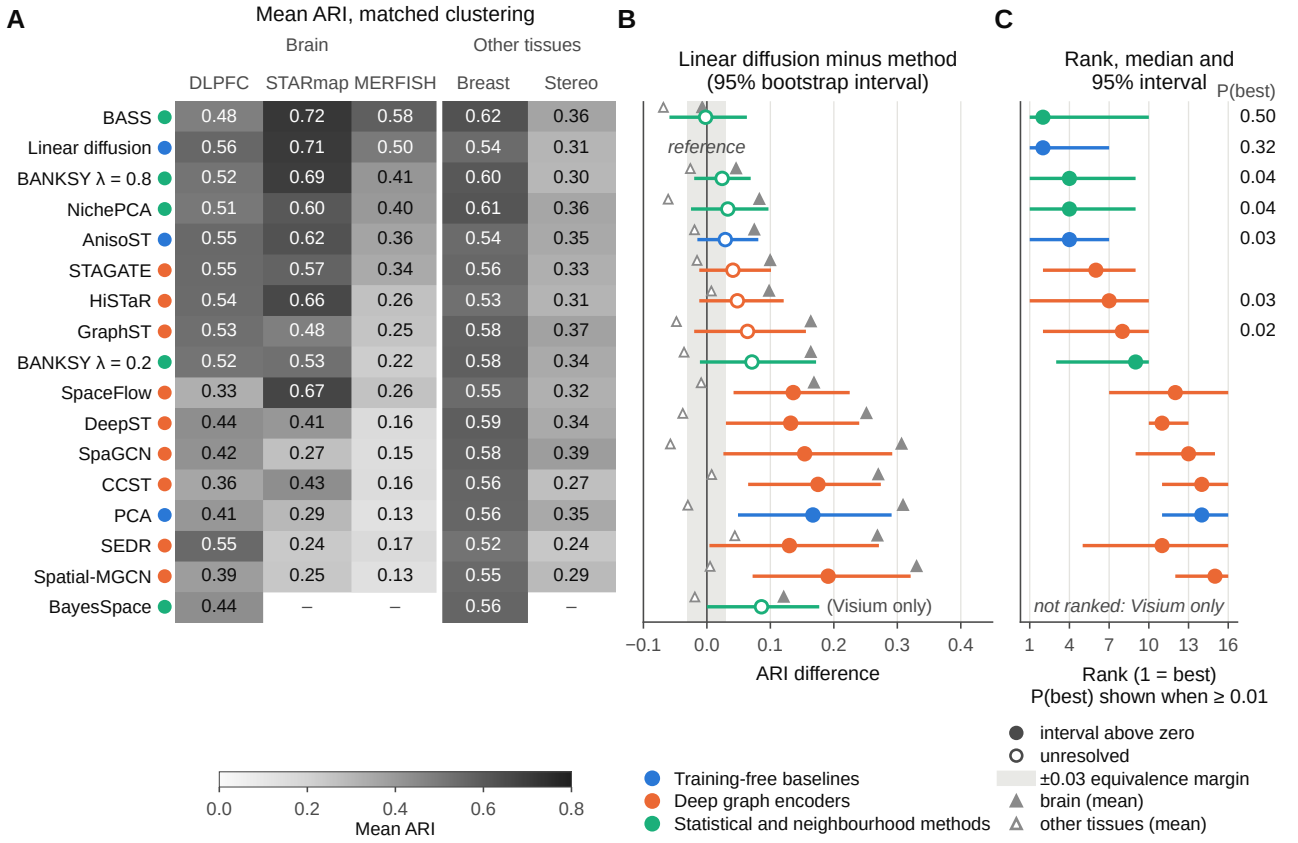


Fig. 2. Accuracy with matched clustering (every embedding clustered with the same 20-start Gaussian mixture; SpaGCN, BASS and BayesSpace use their internal clustering). $n = 24$ sections (12 DLPFC, 3 STARmap, 5 MERFISH, 1 breast cancer, 3 Stereo-seq); values are means over seeds. STARmap: mouse medial prefrontal cortex; MERFISH: mouse hypothalamus; Stereo-seq: mouse embryo. (**A**) Mean ARI per dataset, brain datasets left of the white rule; rows ordered by the mean over datasets, BayesSpace (Visium only) last; dots mark the method family. (**B**) Linear diffusion minus each method, with 95% intervals from a hierarchical bootstrap over 7 groups (three DLPFC donors, STARmap, MERFISH, breast cancer, Stereo-seq; five of the seven groups are brain tissue). Circles and lines: filled, interval above zero; open, unresolved. Small grey marks: mean difference on the three brain datasets (filled triangle) and on breast cancer and Stereo-seq (open triangle), without intervals (per dataset in Supplementary Table S6). Grey band: equivalence margin of ±0.03 ARI. BayesSpace was run on the Visium data only. (**C**) Rank in the bootstrap (dot: median; line: 95% interval; 16 methods ranked); numbers give the probability of ranking first when at least 0.01. Ranks use all 24 sections (BayesSpace, run on the Visium data only, is not ranked). Linear diffusion was chosen as the reference after seeing the means, so this comparison is exploratory.

Figure 4B), and the range exceeded 0.1 for 48% of them; the STARmap sections, with about 1050 cells, were initialised on all cells and did not vary. Initialising on all observations makes the fit deterministic, but it remained sensitive to noise far below the measurement noise: adding 1% noise changed ARI by 0.08 on average (up to 0.59; on the 18,408-bin Stereo-seq section, where one full-data fit takes about 30 min, only for three embeddings). Among embeddings with more than 2000 observations, the full-data fit fell outside the range of the ten seeded fits for 16% (Supplementary Table S12).

Selecting fits by likelihood helps on cortex but not on the other tissues

A natural fix is to run many starts and keep the most likely fit. On the same embedding and model, the best-likelihood fit beat the average single start by more than 0.005 ARI for 10 of 14 methods and was worse for 3 (Supplementary Table S11). Replacing each method's default clustering by matched clustering changed mean ARI by more than 0.005 upwards for 8 of 14 methods and downwards for 5 (Figure 5; Supplementary Table S13); for example HiSTaR gained +0.042 ($p = 0.006$) and NichePCA +0.078 ($p = 0.018$), whereas SEDR changed by −0.077 ($p = 0.074$) and CCST by −0.059 ($p = 0.010$), all uncorrected. The reason is visible in Figure 4C: likelihood and ARI were positively related on DLPFC (median Spearman ρ 0.38) and STARmap (0.41) but not on MERFISH (median ρ −0.11; negative for 10 of 14 methods), breast cancer (−0.13) or Stereo-seq (0.06). On MERFISH the best-fitting mixture may separate cell types rather than anatomical regions. Better optimisation of the mixture model is therefore not the same as better domains, and the clustering protocol alone can reorder methods.

HiSTaR's published DLPFC accuracy was not reproduced

With our PyTorch port of the official code and its published defaults, HiSTaR reached a median ARI over the twelve DLPFC sections of 0.509 (mean 0.491, 95% bootstrap interval over sections 0.454–0.525) with the 3-start GMM, and 0.449 (mean, seed 0) when its embedding was clustered with R mclust as in the official pipeline, compared with a reported median of 0.65 over the twelve sections and 0.718 on section 151673 [10].

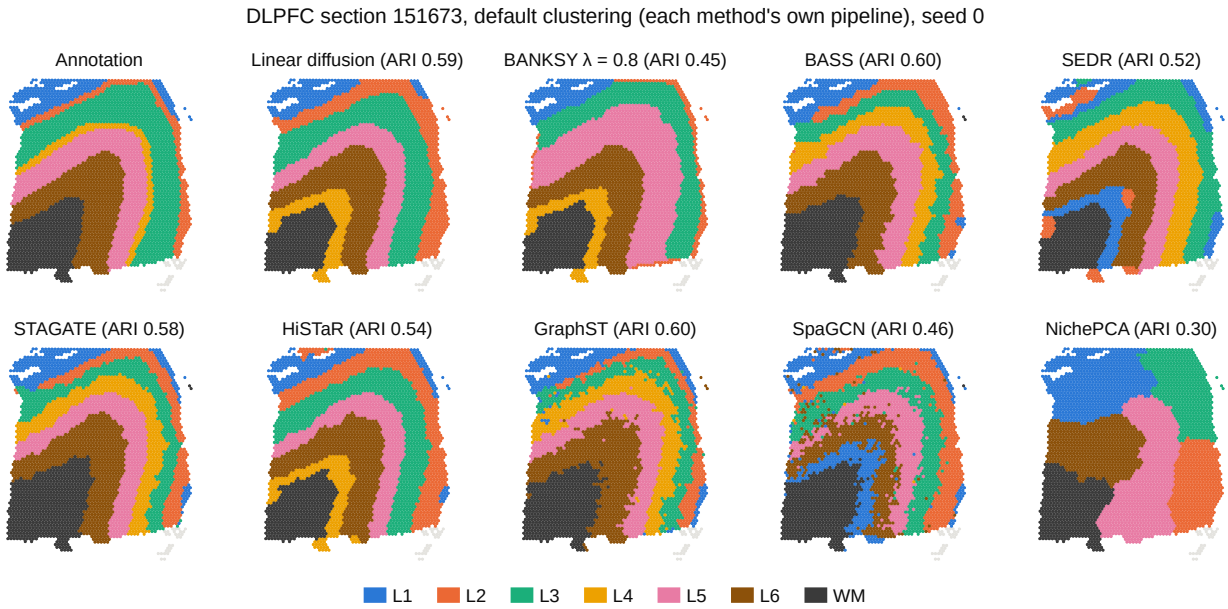


Fig. 3. Domains on DLPFC section 151673 (donor 3, not used for development; seed 0, default clustering in the standardised runner; GraphST without label refinement). The manual annotation and the partitions of nine methods spanning the accuracy range; predicted clusters were matched to the annotated layers with the Hungarian algorithm for display only, so a colour means the matched layer, and the ARI is given in each title. Maps for breast cancer and Stereo-seq are in Supplementary Figure S3. Spots without annotation are light grey.

Fixing a bug that applies random masking at inference did not change accuracy (0.487; Supplementary Table S19). Possible causes include per-section tuning of the loss weights (described in the HiSTaR supplement), unpublished preprocessing, the shared PCA input of this benchmark (our port received the same principal components as the other methods) and the choice of seed; the results above show that the clustering step alone can produce per-section differences of this size. We report these numbers to document what the released code and defaults reproduce, not to question the method itself.

Released code of one method selects the reported epoch using the labels

Spatial-MGCN's released test scripts (`DLPFC_test.py`, `HBC_test.py`) compute ARI against the annotation at every training epoch and keep the best epoch for reporting. Applying the same selection rule in our runs (seed 0, 20 sections), this label-selected epoch had a mean ARI of 0.551 on DLPFC, 0.687 on STARmap and 0.221 on MERFISH, against 0.418, 0.303 and 0.182 for the final epoch used in our benchmark, an inflation of 0.147 on average (0.38 on STARmap; Supplementary Table S20). We do not know whether the published numbers were obtained with these scripts; we report what the released code does. Selecting a run, epoch or seed by agreement with the labels is a form of label leakage: on DLPFC it makes the method look competitive with the training-free baselines (0.551 against 0.558) when its label-free result is not. On the breast cancer section, for which the released repository provides a configuration, the inflation was smaller (0.03 ARI over 3 seeds). An automated screen of the released repositories (11 benchmarked methods and three further spatial methods) for other patterns of label use, with every candidate line read by hand, is reported in Supplementary Table S21 and Supplementary Note 5.

Discussion

We benchmarked 13 published methods and three training-free baselines on 24 sections from four technologies and three tissue types and separated the learned embedding from the clustering step. When all embeddings were clustered in the same way, linear diffusion had a higher mean ARI than 8 of 16 comparators and could not be separated from the rest; the intervals were too wide to show equivalence with any method, and BASS ranked first more often (0.50 against 0.32). This lead came from the brain datasets: on breast cancer and mouse embryo the baseline was slightly behind most methods, including PCA without smoothing, and the methods were close to each other. The clustering step was a larger source of variation than most differences between methods: random initialisation of the same mixture model spanned 0.16 ARI on a fixed embedding, the default fit of R mclust, the final step of GraphST, STAGATE and SEDR, depended on a fixed but arbitrary random seed, and better likelihood did not reliably mean better domains. Reproducibility was limited both by the clustering step and by choices in released code, including selecting a reported epoch with the labels.

Why does simple smoothing do so well on brain tissue? On layered tissue such as cortex, domains are large, contiguous regions, and averaging each spot with its neighbours removes most of the sampling noise that separates spots of the same layer; much of what a graph encoder learns may be this smoothing. In tumour and embryo sections, domains are smaller, interleaved and defined partly by cell composition; a possible explanation, which we did not test, is that a fixed amount of smoothing tuned on cortex then blurs them. There the methods performed similarly, and smoothing gave no gain over PCA alone. The ranking of methods is therefore tissue-dependent, and a benchmark on one tissue, especially DLPFC alone, does not identify a generally best method. On single-cell

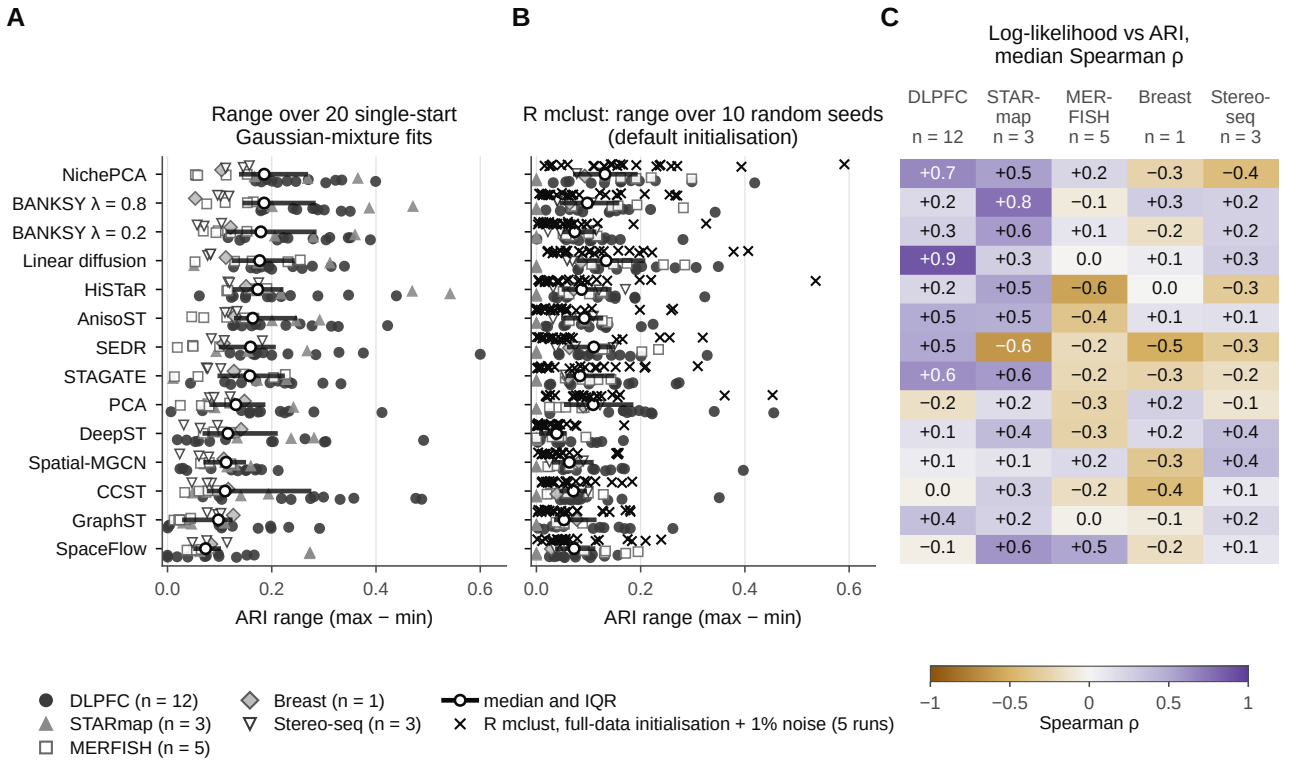


Fig. 4. Variance of the clustering step on fixed embeddings (14 methods and baselines that output an embedding; one symbol per section, up to 24 per method; shape and grey level: dataset). (**A**) ARI range over 20 single-start fits of the same Gaussian mixture. (**B**) ARI range of R mclust with its default initialisation over ten R random seeds; crosses: range over five repetitions with 1% noise when initialised on all observations. STARmap sections have fewer than 2000 cells, are initialised on all cells and show no seed variation (their zero ranges are included in the medians). In A and B, bars: interquartile range; open circle: median; rows are ordered by the median range in A, and C uses the same rows. (**C**) Median Spearman correlation between the log-likelihood and the ARI of the 20 fits in A (positive: better-fitting mixtures agree better with the annotation); columns rest on 12, 3, 5, 1 and 3 sections.

MERFISH data, neighbouring cells often belong to different cell types, and the best-fitting mixture can then describe cell types rather than regions, which may explain why likelihood and ARI disagree there. For a practitioner, the results suggest starting with a smoothing baseline, choosing the clustering step deliberately and reporting its variance. These results agree with recent benchmarks [17; 16; 15] and add a direct measurement of clustering variance on fixed embeddings for 14 methods across four technologies.

Recommendations.

(1) Report the clustering protocol: algorithm, implementation and version, initialisation, number of starts, random seeds and how they were chosen (Figure 4A,B). (2) Never select seeds, epochs or runs by agreement with the labels (Supplementary Table S20). (3) Report distributions over training seeds and over clustering runs or R seeds, with per-section values (Supplementary Tables S10–S12). (4) Compare encoders under the same clustering, include training-free baselines such as neighbourhood smoothing, and evaluate on more than one tissue (Figure 2A, Figure 5). (5) Report run time, memory and seed variance alongside accuracy (Supplementary Table S15). (6) Release per-run results and seeds so that variance can be recomputed. (7) If fits are selected, use a label-free criterion and report how often it agrees with the annotation.

Limitations.

All methods were run with default or published settings, without a tuning budget, so the results do not show what each method can reach with its authors' own tuning (we could not reproduce HiSTaR's per-section tuning). The reference baseline was chosen after the means were seen, and the comparisons with it are exploratory. Four methods (and GraphST on DLPFC) were run with one seed in the published pipelines and BayesSpace on Visium only (Supplementary Table S23), so seed variance on the first three datasets is unavailable for them. Five methods needed more than the 12 GB of the benchmark machine on the largest Stereo-seq section (up to 28 GB) and were rerun on a server with 125 GB RAM; those runs used a different CPU, so their run times are not compared with the rest, and the memory figures describe these implementations on this section, not the methods in general. NichePCA's default workflow aggregates all genes before PCA; its package also offers other orders of these steps. Settings were the same on every dataset by design; dataset-specific tutorial settings (for example SEDR's Stereo-seq graph or Spatial-MGCN's configuration for this breast cancer section) were not used, and neighbourhood radii were expressed in spot or bin spacings, which on Stereo-seq bin50 data covers a smaller physical area than on Visium; on the square bin grid a six-nearest-neighbour graph also has ties among equidistant bins. The initialisation and R mclust experiments use one seed-0

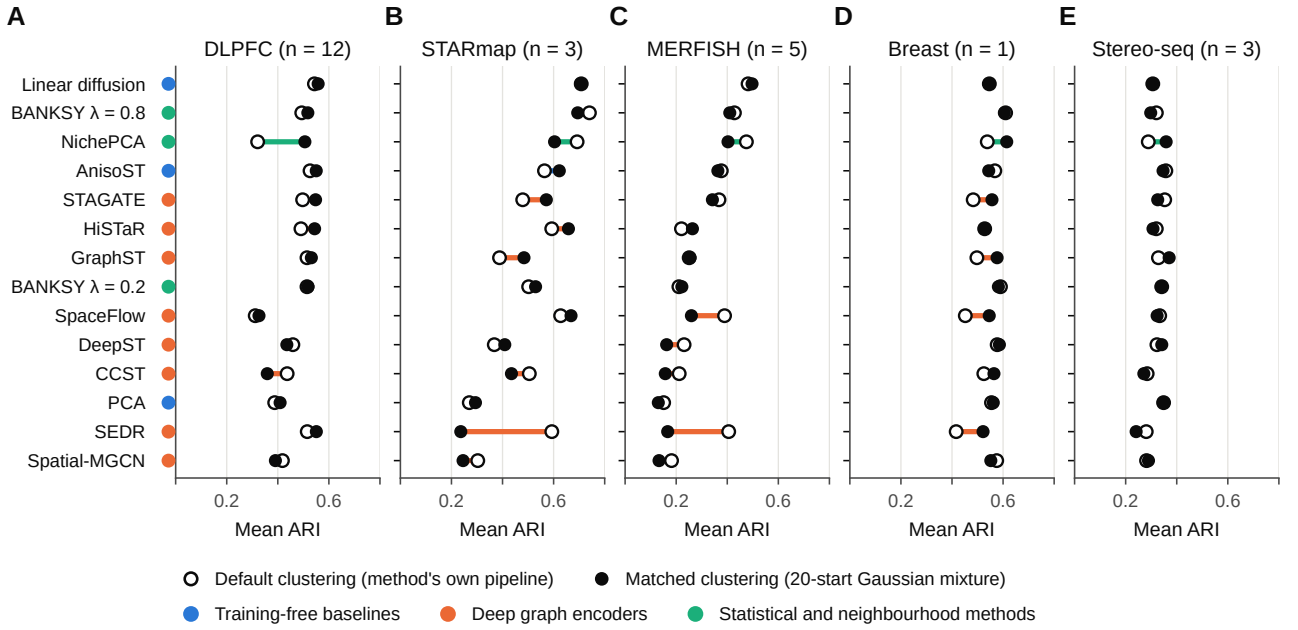


Fig. 5. Effect of the clustering protocol on mean ARI: (**A**) DLPFC, (**B**) STARmap, (**C**) MERFISH, (**D**) breast cancer, (**E**) Stereo-seq (means over 12, 3, 5, 1 and 3 sections). Open circles: default clustering; filled circles: matched clustering of the same embedding; line colour: method family. Default values come from the published pipelines (A–C) and from the standardised runner (D, E). Rows are ordered by the matched-clustering mean over the five datasets. SpaGCN, BASS and BayesSpace cluster internally and are not shown.

embedding per section and method and do not include training-seed variance. Three of the five datasets are brain tissue, and the non-brain data comprise one breast cancer section and three embryo sections of one study; Slide-seq, Xenium and Visium HD data and sections with more than about 20,000 spots are not covered, and no simulated data were used. The STARmap and MERFISH annotations were distributed with the BASS analyses, and BASS was among the most accurate methods on these data. The MOSTA tissue annotation was derived partly from clustering of the expression data and then curated, which may favour methods similar to that clustering, and the breast cancer annotation was distributed with the SEDR analyses; we did not verify either annotation independently. The number of domains was given to every method; results with a misspecified or label-free K are reported in Supplementary Table S17. All runs used one CPU thread, so run times are not representative of GPU use. ARI is the primary metric; other metrics are in Supplementary Table S14.

Data availability

DLPFC: spatialLIBD (count matrices; `http://spatial.libd.org/spatialLIBD`) and the SEDR analyses repository (layer annotations; `https://github.com/JinmiaoChenLab/SEDR_analyses`) [11]. STARmap and MERFISH with domain annotations: BASS-Analysis repository (`https://github.com/zhengli09/BASS-Analysis`) [22], from the original studies [20; 21]. Breast cancer: 10x Genomics dataset V1_Breast_Cancer_Block_A_Section_1 (`https://www.10xgenomics.com/datasets/human-breast-cancer-block-a-section-1-1-standard-1-1-0`) [23], annotation from the SEDR analyses repository [24]. Stereo-seq: MOSTA atlas, CNGB STOmics database STDS0000058 (`https://db.cngb.org/stomics/mosta/`) [25]. Exact file URLs, repository commits and SHA-256 checksums are listed in `configs/datasets.yaml` and `data/CHECKSUMS.sha256`. Code, download scripts with checksums, per-run results and the scripts that generate every number, table and figure: [TODO: GitHub URL and Zenodo DOI].

Supplementary data

Supplementary Data are available at NAR Online.

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Conflict of interest statement

The authors developed AnisoST, one of the training-free baselines, and earlier HiCAST, a modification of HiSTaR that did not improve on it and is not part of the benchmark. No other conflict of interest is declared.

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